BA UNIT - 4

Market Analytics, Models and metrics, Market Insight – Market data sources, sizing, PESTLE trend analysis, and porter 5 forces analysis – Market basket Analysis, Text Analytics, Spreadsheet Modelling – Sales Analytics: E Commerce sales mode, sales metrics profitability metrics and support metrics.

<u>What is Marketing Analytics?</u>

Marketing analytics is the practice of using data to evaluate the effectiveness and success of marketing activities. Marketing analytics allows you to gather deeper consumer insights, optimize your marketing objectives, and get a better return on investment.

Marketing analytics benefits both marketers and consumers. This analysis allows marketers to achieve higher ROI on marketing investments by understanding what is successful in driving either conversions, brand awareness, or both. Analytics also ensures that consumers see a greater number of targeted, personalized ads that speak to their specific needs and interests, rather than mass communications that tend to annoy.

Marketing data can be analyzed using a variety of methods and models depending on the KPIs being measured. For example, analysis of brand awareness relies upon different data and models than analysis of conversions. Some popular analytics models and methods include:

Media Mix Models (MMM): Attribution models that look at aggregate data over a long period of time. Media mix modeling (MMM), sometimes referred to as marketing mix modeling, is an analysis technique that allows marketers to measure the impact of their marketing and advertising campaigns to determine how various elements contribute to their goal, which is often to drive conversions. The insights derived from media mix modeling allow marketers to refine their campaigns based on a variety of factors, ranging from consumer trends to external influencers, to ultimately create an ideal campaign that will drive engagement and sales.

The statistical analysis performed by media mix modeling uses multi-linear regression to determine the relationship between the dependent variable, such as sales or engagements, and the independent variables, such as ad spend across channels. For example, MMM can use both linear and non-linear regression methods to determine how increased marketing spend on magazine ads affected overall sales. To get the most robust and accurate visibility into marketing impact, several models should be evaluated.

Multi-Touch Attribution (MTA): Attribution models that provide person-level data from across the buyer's journey.

Very few customers buy on their first visit to your website. They usually see a few ads, read some blogs, and hear about you from friends. As a result, multi-channel attribution has become critically important to spot nascent and high-quality user acquisition channels

Multi-touch attribution is a marketing effectiveness measurement technique that takes all of the touchpoints on the consumer journey into consideration and assigns fractional credit to each so that a marketer can see how much influence each channel has on a sale.

Unified Marketing Measurement (UMM): A form of measurement that integrates various attribution models including MMM and MTA into comprehensive engagement metrics.

Unified Marketing Measurement means bringing together two of the most important components of marketing analysis. This means Media or Marketing Mix Modeling (MMM) and Multi-Touch Attribution (MTA). Incidentally, UMM is also often referred to as Unified Marketing Impact Analytics (UMIA). The goal of UMM is not to apply MMM and MTA simultaneously. Rather, the aim is to create a uniform model for evaluating marketing effectiveness. Better still it is about giving marketers a holistic view with consistent data to get even closer to the so-called marketing truth. In concrete terms, this means that with UMM, the aggregated data from the Marketing Mix Modeling must be merged with the user-level data from the Multi-Touch Attribution.

What can you do with marketing analytics?

You can answer the following questions with marketing analytis:

How are our marketing activities performing today?
How about in the long run?
What can we do to improve them?
How do our marketing activities compare with our competitors?
Are they using channels that we aren't using?
What should we do next?
Are our marketing resources properly allocated?
Are we devoting time and money to the right channels?
How should we prioritize our investments over a certain time period?

How marketing analytics works

Marketing analytics requires more than just flashy tools. Marketing teams need a strategy that puts all their data in perspective. Here's how marketing analytics works for most organizations.

1. Identify what you want to measure

Define exactly what you're hoping to accomplish through your marketing. Start with the overall goal of your marketing strategy, then start drilling down into specific campaigns and marketing channels. Metrics can include return on investment, conversion rate, click rate or brand recognition. You also want to define benchmarks and milestones along the way that will help you evaluate and adapt your marketing techniques.

2. Use a balanced assortment of analytic techniques and tools

To get the most benefit from marketing analytics, you'll want a balanced assortment of techniques and tools. Use analytics to:

Report on the past: By using techniques that look at the past, you can answer questions such as: What campaign elements generated the most revenue last quarter? How did social media campaign A perform against direct mail campaign B? How many leads did we generate from this webinar series vs.

that podcast season?

Analyze the present: Determine how your marketing initiatives are performing right now. How are customers engaging with us? Which channels do our most profitable customers prefer? Who is talking about us and where?

Predict or influence the future: Marketing analytics can deliver data-driven predictions that help you shape the future. You can answer questions such as: How can short-term wins be molded into loyalty? How will adding more sales representatives in underperforming regions affect revenue? Which cities should we target next?

3. Assess your analytic capabilities, and fill in the gaps

Marketing analytics technology is abundant so it can be hard to know which tools you really need. But don't start there; start with your overall capability. Assess your current capabilities to determine where you are along the analytics spectrum. Then start identifying where the gaps are and develop a strategy for filling them in.

4. Act on what you learn

Using data is one of the greatest challenges facing marketing professionals these days. There's just so. Much. Data! That's why Step 1 is so important: If you know that what you're currently doing isn't helping you reach your goals, then you know it's time to test and iterate.

Applied holistically, marketing analytics allows for more successful marketing campaigns and a better overall customer experience. Specifically, when acted upon, marketing analytics can lead to better supply and demand planning, price optimization, and robust lead nurturing and management – all of which leads to greater profitability.

Market Analytics Models

Customer acquisition modelling

Marketing is not just about finding an audience - it's about finding the right audience. Customer acquisition modelling revolves around identifying potential prospects who are likely to be "good customers", as well as discovering traits that drive a higher likelihood of a population becoming customers. This allows marketers to better target their customer acquisition efforts, improving overall efficiency and ROI.

Example: A non-profit organisation has gathered a significant amount of personal information on potential donors through its website but is unsure which of these are most likely to convert into full donors. It uses a highly complex customer acquisition model, training it on historical data. The model provides a ranked list of contacts most likely to convert. Marketers then reach out to these specific contacts via email with a simple request to make a recurring donation each month.

Recommendation modelling

Recommendation modelling encompasses a suite of models designed to identify opportunities to improve the overall value of a given customer’s relationship through higher value or more diverse purchases. It includes upsell modelling, cross-sell modelling and next best offer modelling. The major differences between these models are the desired business outcomes:

Upsell modelling deals with increasing purchase value. It relies on knowing which products are most valuable to the business and predicting and identifying which customers are most likely to purchase those products.
Example: A retailer feeds customer information into a highly complex upsell model, which then ranks the customers most likely to purchase higher-value products. Marketers reach out to these customers individually, offering a discount on their next purchase of a specified high-profitability item.

Cross-sell modelling deals with increasing purchase diversity. It relies on finding those customers who are most likely going to expand their purchasing behaviour to another subset of offerings. Example: A marketing firm feeds customer information into a highly complex cross-sell model, which then ranks the customers most likely to expand their product purchasing behaviour. Marketers reach out to these customers, offering a free trial of any of their other services.

Next best offer modelling deals with encouraging overall purchasing value across the entire catalogue of offers. It relies on identifying which products or services are most likely to appeal to a set of customers and is generally more focused on customer needs.
Example: A bank feeds customer information into a highly complex next best offer model, which associates customers with specific products they are likely to purchase. Marketers design a set of push notifications to be deployed through the online banking function to these customers, letting them know about the existence and benefits of their individually associated products.

Fast-track modelling

Fast-track modelling predicts which customers are most likely to become high-value clients over time.

This is different to merely identifying high-potential customers overall, as it takes the influence of tenure into account as well - a cohort of new customers might not be high value now, but they could become so after a year or two. Once identified, these customers are then prioritised by marketing efforts designed to push them further down the value cycle.

Example: A credit card provider feeds the information of every new customer into a highly complex fast track model, which determines the likelihood of this new customer eventually becoming high value. Should the new customer breach a given threshold of likelihood, they are set as a priority for marketing efforts. They are sent offers for lower interest rates over a fixed term sooner than other customers, encouraging them to make more regular, higher-value purchases – a behaviour which should persist after the fixed term concludes.

Churn modelling

Churn modelling is used to identify customers who are likely to stop doing business with a given company within a short amount of time. It is often used as a form of an early warning detection system, flagging high-risk customers that require outreach. An important point to note is that churn modelling is mostly used for companies that operate on a customer 'binary' - either a person is their customer, or they're not. Power companies are a good example, as are telecommunications businesses (barring the minority of users who have more than one phone or internet connection).

Example: A power company uses churn modelling to identify current customers who are at a high risk of churning. Once flagged, these customers are prioritised for outreach marketing material, highlighting the benefits of staying with said company and offering a loyalty discount.

Slider modelling

Slider modelling is used to predict which customers are likely to "slide" towards doing business with a competitor, reducing a business' share of wallet. It is similar to churn modelling, but where churn modelling deals with the 'customer binary', slider modelling is less discreet. It is the difference between discovering which customers are going to stop doing business with a company altogether (churn) compared to them preferring to do business with a competitor – or no one at all (slide). Companies that can 'share' customers with a competitor typically use slider modelling rather than churn modelling. Retailers are a common example.

Example: A petrol company uses slider modelling to identify which customers are likely to start spending less at their petrol stations and (presumably) more at competitors. The marketers reach out to these customers with fuel coupons to entice them back before their slide continues.

Customer commitment modelling

Customer commitment modelling aims to understand who the most loyal (and least loyal) customers are in an existing client base and rank them accordingly. It may also involve identifying the behaviours that are associated with higher levels of loyalty. This information can then be used to differentiate marketing messages depending on customer commitment, targeting the disposition and likelihood of each given segment more specifically. It could also be used to reward customers who are the most loyal, encouraging higher lifetime value and more word of mouth marketing.

Example: A bank uses customer commitment modelling to identify customers that are most likely to have higher levels of satisfaction with its products and services. Marketers reach out to these customers with requests for testimonials, using these positive social proofs on their website to convert new visitors.

Marketing Metrics You Need To Measure

When establishing your digital marketing analytics dashboard, there are core metrics you should include to help showcase the ROI of your full-funnel initiatives at a glance.

10 of the most important marketing metrics include:

Traffic by source
Unique monthly visitors
Cost per lead (CPL)
Cost per acquisition (CPA)
Conversion rates by source and device
Return on advertising spend (ROAS)
Page performance
Customer lifetime value (CLV)
Average Search Engine Results Page (SERP) Position
Branded search queries over time

Market data sources

Market Sizing Techniques

Market Sizing is a method of evaluating the potential reach and revenue of your product/service. Market Sizing is Divided broadly into 4 Categories:

1. Market Potential:
Assuming 100% market Share, the total potential value of product/service sold over a specified timeframe.

2. TAM (Total Addressable Market):
The potential value of Product/Service sold to a particular customer segment.

3. SAM(Serviceable Available Market):
The total value of Product/Service that can fulfill the demand of a segment using a one revenue stream/channel.

4. SOM(Serviceable Obtainable Market):
The total value of SAM divided by the expected percentage market share that the company can capture.

<u>Sizing - Why is it Important?</u>

- Quantify the sales or profit potential of new product markets or customer segments.
- Identify product lines and customer segments with lucrative growth opportunities.
- Helps validate Business Model hypothesis.
- Specify competitive threats and develop strategic responses to those threats.
- Develop exit strategies or pivot points for the future.
- Important to Investors: They want a scalable product addressing a large enough market to get the

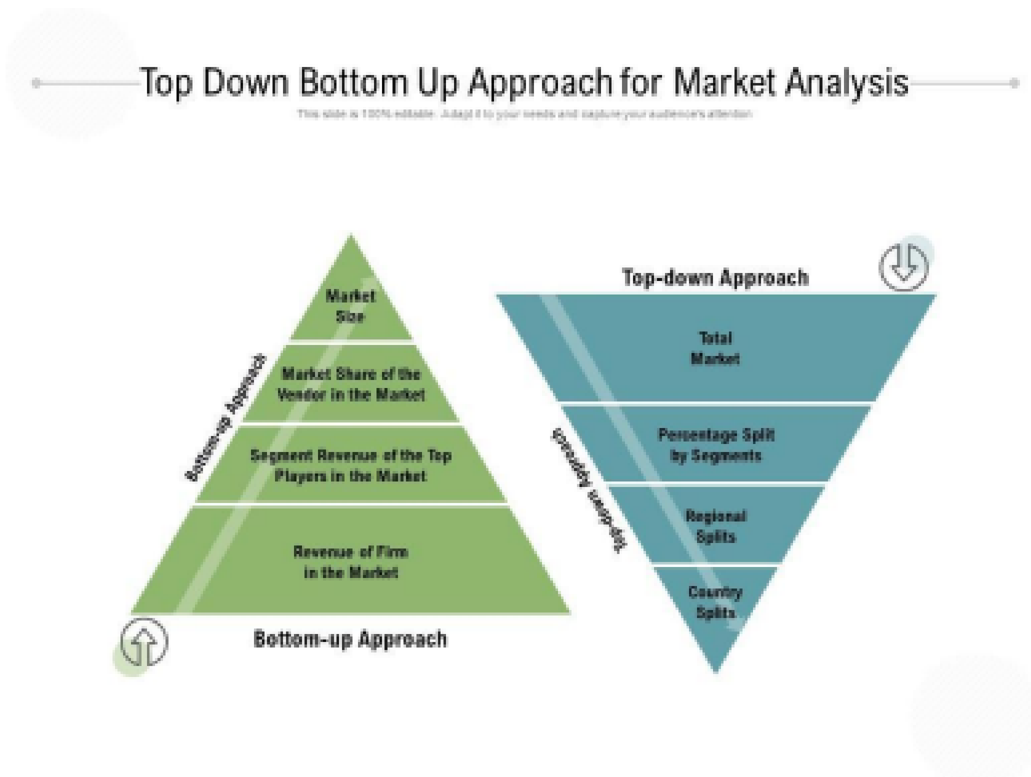
most bang for their bucks. Listen to them.

How to use Market Sizing:

- First, let's ascertain the Quantifiable Values that we are looking for:
- Value: Total value of clients/products in the market.
- Units: Total Number of clients/products in the market
- Market Share: % of clients/products acquired/sold by your competitors vs total value/units.

Market Sizing Approaches:

Top Down: This type of sizing is done generally by using Demographical data like Company size, industry, location or Human Population, Age, Income etc. Base your hypothesis on a large number and work your way down from there. You can segment the market the way it best suits the business. **Bottom Up:** Identify the Micro segment or the niche market you are targeting and extrapolate the numbers till you reach an appropriate scale. Eg: Calculate how much an average person spends a day in the office cafeteria, and now multiple it by the number of employees and number of working days in a month/year.

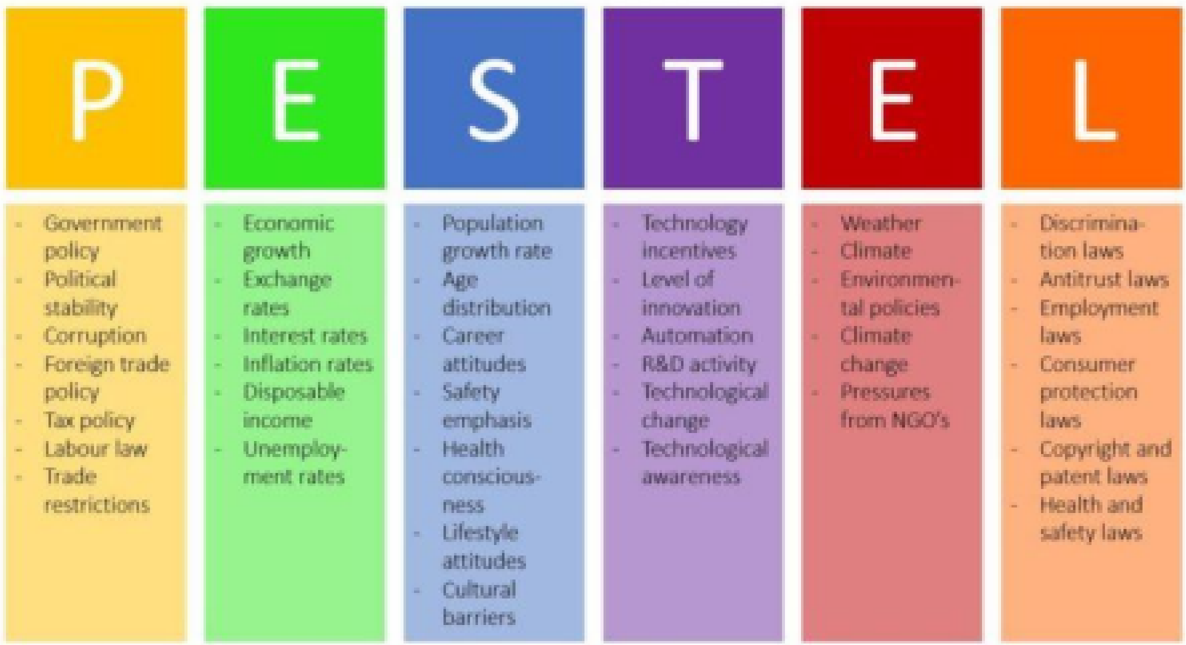


For both approaches, include price at the last step. We recommend using both methods to try to solve the problem. Both approaches might produce different results, but in general, they should be close enough to triangulate on a more accurate answer.

<u>What is PESTLE Analysis? An Important Business Analysis Tool</u>

What is PESTLE Analysis? PESTLE analysis, which is sometimes referred to as PEST analysis, is a concept in marketing principles. Moreover, this concept is used as a tool by companies to track the environment they're operating in or are planning to launch a new project/product/service, etc. So, let's find out what these letters stand for first.

PESTLE is a mnemonic which in its expanded form denotes P for Political, E for Economic, S for Social, T for Technological, L for Legal, and E for Environmental. It gives a bird's eye view of the whole environment from many different angles that one wants to check and keep a track of while contemplating a certain idea/plan. The PESTEL framework has undergone certain alterations, as gurus of Marketing have added certain things like an E for Ethics to instill the element of demographics while utilizing the framework while researching the market.



Asking the Right Questions

There are certain questions that one needs to ask while conducting this analysis, which gives them an idea of what things to keep in mind. They are:

• What is the political situation of the country and how can it affect the industry? • What are the prevalent economic factors?

• How much importance does culture have in the market and what are its determinants? • What technological innovations are likely to pop up and affect the market structure? • Are there any current legislations that regulate the industry or can there be any change in the legislations for the industry?

• What are the environmental concerns for the industry?

All the aspects of this technique are crucial for any industry a business might be in. More than just understanding the market, this framework represents one of the vertebras of the backbone of strategic management that not only defines what a company should do but also accounts for an organization's goals and the strategies stringed to them. It may be so, that the importance of each of the factors may be different to different kinds of industries, but it is imperative to any strategy a company wants to develop that they conduct the PESTLE analysis as it forms a much more comprehensive version of the SWOT analysis. If you don't know what is a SWOT analysis, we have you covered. It is very critical for one to understand the complete depth of each of the letters of the PESTLE.

Political factors in PESTLE Analysis

These factors determine the extent to which a government may influence the economy or a certain industry. For example, a government may impose a new tax or duty due to which entire revenue generating structures of organizations might change. Political factors include tax policies, Fiscal policy, trade tariffs, etc. that a government may levy around the fiscal year and it may affect the business environment (economic environment) to a great extent.

Economic factors in PESTLE Analysis

These factors are determinants of an economy's performance that directly impacts a company and have resonating long term effects. For example, a rise in the inflation rate of any economy would affect the way companies price their products and services. Adding to that, it would affect the purchasing power of a consumer and change demand/supply models for that economy. Economic factors include inflation rate, interest rates, foreign exchange rates, economic growth patterns, etc. It also accounts for the FDI (foreign direct investment) depending on certain specific industries who're undergoing this analysis.

Social factors in PESTLE Analysis

These factors scrutinize the social environment of the market, and gauge determinants like cultural trends, demographics, population analytics, etc. An example of this can be buying trends for Western countries like the US where there is high demand during the Holiday season.

Technological factors in PESTLE Analysis

These factors pertain to innovations in technology that may affect the operations of the industry and the market favorably or unfavorably. This refers to automation, research and development, and the amount of technological awareness that a market possesses.

Legal factors in PESTLE Analysis

These factors have both external and internal sides. There are certain laws that affect the business environment in a certain country while there are certain policies that companies maintain for themselves. Legal analysis takes into account both of these angles and then charts out the strategies in light of these legislations. For example, consumer laws, safety standards, labor laws, etc.

Environmental factors in PESTLE Analysis

These factors include all those that influence or are determined by the surrounding environment. This aspect of the PESTLE is crucial for certain industries particularly for example tourism, farming, agriculture, etc. Factors of a business environmental analysis include but are not limited to climate, weather, geographical location, global changes in climate, environmental offsets, etc.

There are many templates available for companies to conduct PESTLE analysis. Many organizations have provided information regarding their PESTLE analysis as case studies available on the Internet.

Porter's Five Forces

Porter's Five Forces is a model that identifies and analyzes five competitive forces that shape every industry and helps determine an industry's weaknesses and strengths. Five Forces analysis is frequently used to identify an industry's structure to determine corporate strategy.

Porter's model can be applied to any segment of the economy to understand the level of competition within the industry and enhance a company's long-term profitability. The Five Forces model is named after Harvard Business School professor, Michael E. Porter.

Porter's 5 forces are:

- Competition in the industry
- Potential of new entrants into the industry
- Power of suppliers
- Power of customers
- Threat of substitute products

KEY TAKEAWAYS

Porter's Five Forces is a framework for analyzing a company's competitive environment. Porter's Five Forces is a frequently used guideline for evaluating the competitive forces that influence a variety of business sectors. It was created by Harvard Business School professor Michael E. Porter in 1979 and has since become an important tool for managers. These forces include the number and power of a company's competitive rivals, potential new market entrants, suppliers, customers, and substitute products that influence a company's profitability. Five Forces analysis can be used to guide business strategy to increase competitive advantage.

Understanding Porter's Five Forces

Porter's Five Forces is a business analysis model that helps to explain why various industries are able to sustain different levels of profitability. The Five Forces model is widely used to analyze the industry structure of a company as well as its corporate strategy. Porter identified five undeniable forces that play a part in shaping every market and industry in the world, with some caveats. The Five Forces are frequently used to measure competition intensity, attractiveness, and profitability of an industry or market.

1. Competition in the Industry

The first of the Five Forces refers to the number of competitors and their ability to undercut a company. The larger the number of competitors, along with the number of equivalent products and services they offer, the lesser the power of a company. Suppliers and buyers seek out a company's competition if they are able to offer a better deal or lower prices. Conversely, when competitive rivalry is low, a company has greater power to charge higher prices and set the terms of deals to achieve higher sales and profits.

2. Potential of New Entrants Into an Industry

A company's power is also affected by the force of new entrants into its market. The less time and money it costs for a competitor to enter a company's market and be an effective competitor, the more an established company's position could be significantly weakened. An industry with strong barriers to entry is ideal for existing companies within that industry since the company would be able to charge higher prices and negotiate better terms.

3. Power of Suppliers

The next factor in the Porter model addresses how easily suppliers can drive up the cost of inputs. It is affected by the number of suppliers of key inputs of a good or service, how unique these inputs are, and how much it would cost a company to switch to another supplier. The fewer suppliers to an industry, the more a company would depend on a supplier. As a result, the supplier has more power and can drive up input costs and push for other advantages in trade. On the other hand, when there are many suppliers or low switching costs between rival suppliers, a company can keep its input costs lower and enhance its profits.

4. Power of Customers

The ability that customers have to drive prices lower or their level of power is one of the Five Forces. It is affected by how many buyers or customers a company has, how significant each customer is, and how much it would cost a company to find new customers or markets for its output. A smaller and more powerful client base means that each customer has more power to negotiate for lower prices and better deals. A company that has many, smaller, independent customers will have an easier time charging higher prices to increase profitability.

5. Threat of Substitutes

The last of the Five Forces focuses on substitutes. Substitute goods or services that can be used in place of a company's products or services pose a threat. Companies that produce goods or services for which there are no close substitutes will have more power to increase prices and lock in favorable terms. When close substitutes are available, customers will have the option to forgo buying a company's product, and a company's power can be weakened.

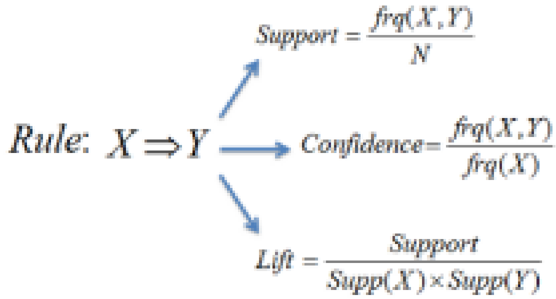
Understanding Porter's Five Forces and how they apply to an industry, can enable a company to adjust its business strategy to better use its resources to generate higher earnings for its investors.

What Are Porter's Five Forces Used for?

Porter's Five Forces Model helps managers and analysts understand the competitive landscape that a company faces and to understand how a company is positioned within it.

<u>Market basket Analysis</u>

Market Basket Analysis is one of the key techniques used by large retailers to uncover associations between items. It works by looking for combinations of items that occur together frequently in transactions. To put it another way, it allows retailers to identify relationships between the items that people buy.



Example:

Rule	Support	Confidence	Lift
$A \Rightarrow D$	2/5	2/3	10/9
$C \Rightarrow A$	2/5	2/4	5/6
$A \Rightarrow C$	2/5	2/3	5/6
$B \& C \Rightarrow D$	1/5	1/3	5/9

Market Basket Analysis in Data Mining

Market basket analysis is a data mining technique used by retailers to increase sales by better understanding customer purchasing patterns. It involves analyzing large data sets, such as purchase history, to reveal product groupings and products that are likely to be purchased together. The adoption of market basket analysis was aided by the advent of electronic point-of-sale (POS) systems. Compared to handwritten records kept by store owners, the digital records generated by POS systems made it easier for applications to process and analyze large volumes of purchase data.

One example is the Shopping Basket Analysis tool in Microsoft Excel, which analyzes transaction data contained in a spreadsheet and performs market basket analysis. A transaction ID must relate to the items to be analyzed. The Shopping Basket Analysis tool then creates two worksheets:

The Shopping Basket Item Groups worksheet, which lists items that are frequently purchased together,
And the Shopping Basket Rules worksheet shows how items are related (For example, purchasers of Product A are likely to buy Product B).

How does Market Basket Analysis Work?

Market Basket Analysis is modelled on Association rule mining, i.e., the IF {}, THEN {} construct. For example, IF a customer buys bread, THEN he is likely to buy butter as well.
Association rules are usually represented as: {Bread} -> {Butter}

Some terminologies to familiarize yourself with Market Basket Analysis are:

Antecedent: Items or 'itemsets' found within the data are antecedents. In simpler words, it's the IF component, written on the left-hand side. In the above example, bread is the antecedent. **Consequent:**A consequent is an item or set of items found in combination with the antecedent. It's the THEN component, written on the right-hand side. In the above example, butter is the consequent.

Types of Market Basket Analysis

Market Basket Analysis techniques can be categorized based on how the available data is utilized. Here are the following types of market basket analysis in data mining, such as:

Descriptive market basket analysis: This type only derives insights from past data and is the most frequently used approach. The analysis here does not make any predictions but rates the association between products using statistical techniques. For those familiar with the basics of Data Analysis, this type of modelling is known as unsupervised learning.

Predictive market basket analysis: This type uses supervised learning models like classification and regression. It essentially aims to mimic the market to analyze what causes what to happen. Essentially, it considers items purchased in a sequence to determine cross-selling. For example, buying an extended warranty is more likely to follow the purchase of an iPhone. While it isn't as widely used as a descriptive MBA, it is still a very valuable tool for marketers.

Differential market basket analysis: This type of analysis is beneficial for competitor analysis. It compares purchase history between stores, between seasons, between two time periods, between different days of the week, etc., to find interesting patterns in consumer behaviour. For example, it can help determine why some users prefer to purchase the same product at the same price on Amazon vs Flipkart. The answer can be that the Amazon reseller has more warehouses and can deliver faster, or maybe something more profound like user experience.

Algorithms associated with Market Basket Analysis

In market basket analysis, association rules are used to predict the likelihood of products being purchased together. Association rules count the frequency of items that occur together, seeking to find associations that occur far more often than expected.
Algorithms that use association rules include AIS, SETM and Apriori. The Apriori algorithm is commonly cited by data scientists in research articles about market basket analysis. It identifies frequent items in the database and then evaluates their frequency as the datasets are expanded to larger sizes.

With the help of the Apriori Algorithm, we can further classify and simplify the item sets that the consumer frequently buys. There are three components in APRIORI ALGORITHM: • SUPPORT

- CONFIDENCE
- LIFT

For example, suppose 5000 transactions have been made through a popular e-Commerce website. Now they want to calculate the support, confidence, and lift for the two products. For example, let's say pen and notebook, out of 5000 transactions, 500 transactions for pen, 700 transactions for notebook, and 1000 transactions for both.

SUPPORT

It has been calculated with the number of transactions divided by the total number of transactions made,

Support = freq(A, B)/N

support(pen) = transactions related to pen/total transactions

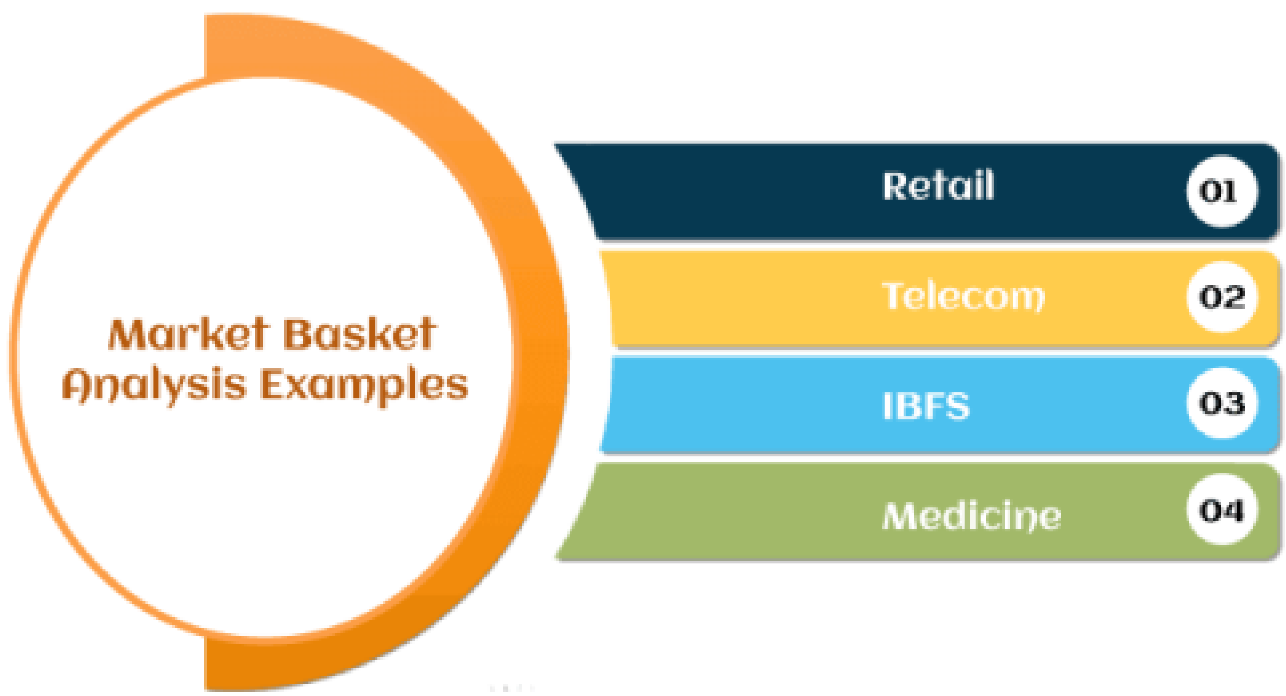
i.e support -> 500/5000=10 percent

CONFIDENCE

Whether the product sales are popular on individual sales or through combined sales has been

calculated. That is calculated with combined transactions/individual transactions.
Confidence = freq (A, B)/ freq(A)
Confidence = combine transactions/individual transactions
i.e confidence-> 1000/500=20 percent
LIFT
Lift is calculated for knowing the ratio for the sales.
Lift = confidence percent/ support percent
Lift-> 20/10=2

When the Lift value is below 1, the combination is not so frequently bought by consumers. But in this case, it shows that the probability of buying both the things together is high when compared to the transaction for the individual items sold.
Examples of Market Basket Analysis
Here are the following examples that *explore Market Basket Analysis by market segment, such as:*



Retail: The most well-known MBA case study is Amazon.com. Whenever you view a product on Amazon, the product page automatically recommends, "Items bought together frequently." It is perhaps the simplest and most clean example of an MBA's (Market Basket Analysis) cross-selling techniques. Apart from e-commerce formats, BA is also widely applicable to the in-store retail segment. Grocery stores pay meticulous attention to product placement based and shelving optimization. For example, you are almost always likely to find shampoo and conditioner placed very close to each other at the grocery store. Walmart's infamous beer and diapers association anecdote is also an example of Market Basket Analysis.
Telecom: With the ever-increasing competition in the telecom sector, companies are paying close attention to customers' services. For example, Telecom has now started to bundle TV and Internet packages apart from other discounted online services to reduce churn.
IBFS: Tracing credit card history is a hugely advantageous MBA opportunity for IBFS organizations. For example, Citibank frequently employs sales personnel at large malls to lure potential customers with attractive discounts on the go. They also associate with apps like Swiggy and Zomato to show customers

many offers they can avail of via purchasing through credit cards. IBFS organizations also use basket analysis to determine fraudulent claims.

Medicine: Basket analysis is used to determine comorbid conditions and symptom analysis in the medical field. It can also help identify which genes or traits are hereditary and which are associated with local environmental effects.

Benefits of Market Basket Analysis

The market basket analysis data mining technique has the following benefits, such as:

Increasing market share: Once a company hits peak growth, it becomes challenging to determine new ways of increasing market share. Market Basket Analysis can be used to put together demographic and gentrification data to determine the location of new stores or geo-targeted ads. **Behaviour analysis:** Understanding customer behaviour patterns is a primal stone in the foundations of marketing. MBA can be used anywhere from a simple catalogue design to UI/UX. **Optimization of in-store operations:** MBA is not only helpful in determining what goes on the shelves but also behind the store. Geographical patterns play a key role in determining the popularity or strength of certain products, and therefore, MBA has been increasingly used to optimize inventory for each store or warehouse.

Campaigns and promotions: Not only is MBA used to determine which products go together but also about which products form keystones in their product line.

Recommendations: OTT platforms like Netflix and Amazon Prime benefit from MBA by understanding what kind of movies people tend to watch frequently.

Text Analytics

Text analytics combines a set of machine learning, statistical and linguistic techniques to process large volumes of unstructured text or text that does not have a predefined format, to derive insights and patterns. It enables businesses, governments, researchers, and media to exploit the enormous content at their disposal for making crucial decisions. Text analytics uses a variety of techniques – sentiment analysis, topic modelling, named entity recognition, term frequency, and event extraction.

Businesses are interacting with customers more than ever. Most businesses feel the pressure to be omnichannel and omnipresent. The digital nature of the relationship also means that every click, every payment, every message can be tracked and measured.

It’s precisely because there is so much information that we struggle to truly know our customers. Information overload & Analysis paralysis. We miss crucial insights because there is simply so much.

Text analysis helps businesses analyse huge quantities of text-based data in a scalable, consistent and unbiased manner. Without the need for excessive resources, it analyses data and extracts valuable information, leaving companies free to action on those insights.

Given 80% of business information is mostly unstructured textual data, this form of intelligent automation has become crucial for the modern enterprise to continue attract, engage and satisfy customers while staying ahead of the competition.

Text Analysis is the process of analysing unstructured and semi-structured text data for valuable insights, trends and patterns. It is typically used in instances where there is a need to process large volumes of text-based data for insights, but would otherwise be too resource and time-intensive to be analysed manually by humans.

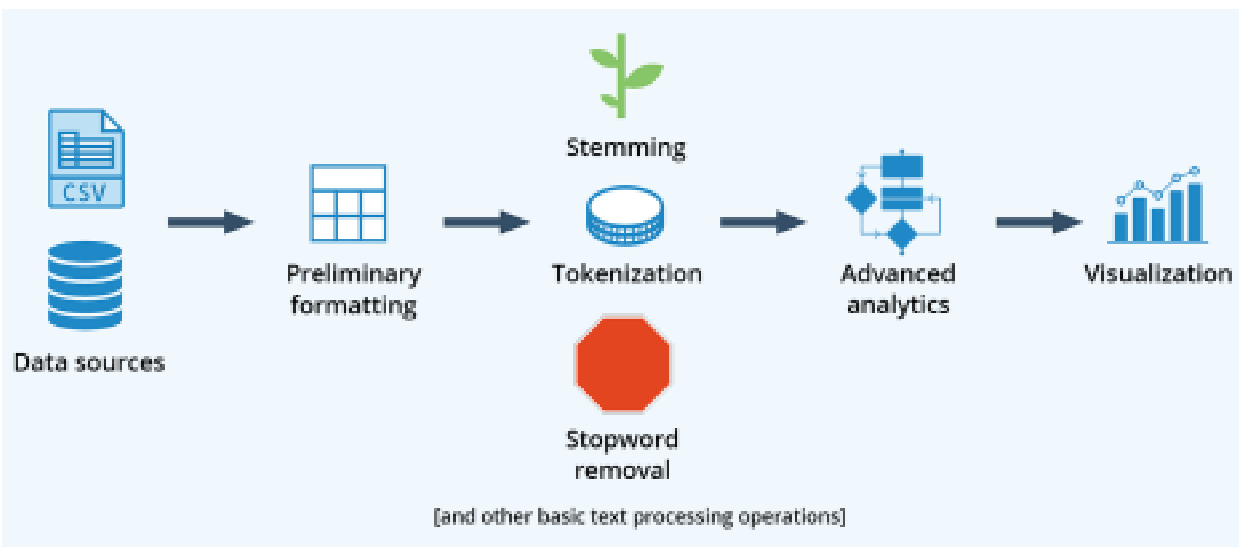
Some instances where text analysis would be used include:

- Looking into the last 3 years of support tickets for trending customer concerns over time
- Analysing 1,000 customer surveys for new product launch feedback
- Finding the most impactful topics to build into your chatbot or knowledge base

In all these cases, text analysis methods would outperform humans. More than just being less time and resource heavy, the final insights are also more consistent with fewer human errors or biases interfering with the process.

As such, many organisations across all industries are using text analysis methods to gain quantitative and qualitative understanding of their text datasets. These are then used in conjunction with data visualisation tools to better translate the information into actionable insights for informed decision making.

Basic Text Processing Pipeline



"At a high level, a text analytics system is a pipeline process comprising three stages:

- Text acquisition and preparation
- Processing and analysis
- Output (that is, visualization, presentation and deployment)."

Text acquisition and preparation is the most straightforward stage. Many text analytics tools can read directly from flat files such as Excel files, relational databases and web sources, e.g., Twitter and Facebook. The text is then prepared using some basic operations shown in the above diagram.

Processing and analysis stage - Text analytics involves "a combination of both linguistic and machine-learning statistical techniques in processing and analysis of text source. Where the linguistic technique is provided for data preparation and customization of the model, the machine-learning technique will be applied to get a more rapid result for weighting. That means the text source will be 'parsed' and 'chunked' linguistically before any run through of the machine-learning algorithms for classification."

Stopword removal: Textual data contains lots of little insignificant yet highly common words: "to," "for," "of," "the" etc. If you're analyzing the frequency of words in a document, these will always surface to the top. Stopword removal is the process of filtering out these words to look for longer, more significant words. Many text analytics solutions come with built-in stopword dictionaries for different languages, and you can also build your own.

Tokenization:"Tokenization looks at the spaces between the words as word boundaries in order to 'cut' words from the text string at those points." This isn't always obvious as it would seem—for instance, is "garbage truck" one word or two? Tokenization is a first step toward structuring text for further analysis, as otherwise text is treated as an unstructured "string." Once the string has been broken into individual words or "tokenized," you can determine word frequency in the document and begin identifying relationships between words.

Businesses will thus need to ensure that common compound words in their operations are properly tokenized for further analysis. Most text-mining tools allow users to customize tokenization rules.

Bag-of-words: This operation breaks down a document into its constituent words, regardless of grammar and order. It's never an end-step in a text mining operation, but rather a first step toward further analysis. Bag-of-words analysis is thus a type of tokenization. The difference is that tokenization operations typically preserve the order of "tokens" in the string, while the bag-of-words model disregards word order to focus on word frequency.

Stemming (a.k.a. "lemmatization"): Words don't just have a single form, but many forms based on their grammatical purpose. For instance, "was," "are," "is" etc. are all forms of the verb "to be." Stemming reduces words in a document to their basic forms, including words that are significant to the company's operations and goals. Once you perform these basic operations, further analytics can be applied to textual data in order to recognize patterns.

Cluster analysis: In the context of text mining, cluster analysis involves classifying words into groups, such that words in a given group are more related to each other than they are to words in other groups. This is done through the application of algorithms such as k-means and x means.

"Maybe you want to segment your data into two groups, five groups etc.—you have to put that in there, and then the algorithm builds around that. In x-means, you don't know what the optimal amount of clusters is, so you specify a minimum of, say, two, and a maximum of, say, 60. The algorithm will then run an analysis and output the optimal number of clusters." Clustering is a powerful way to spot relationships and trends in massive collections of textual data. It can also be a first step toward sentiment analysis (e.g., using text mining to identify consumer emotions), since words relating to brands, products, services and other entities can be grouped via clustering.

Benefits of Text Analytics

There are a range of ways that text analytics can help businesses, organizations, and event social movements:

- Help businesses to understand customer trends, product performance, and service quality. This results in quick decision making, enhancing business intelligence, increased productivity, and cost savings.
- Helps researchers to explore a great deal of pre-existing literature in a short time, extracting what is relevant to their study. This helps in quicker scientific breakthroughs.
- Assists in understanding general trends and opinions in the society, that enable governments and political bodies in decision making.
- Text analytic techniques help search engines and information retrieval systems to improve their performance, thereby providing fast user experiences.
- Refine user content recommendation systems by categorizing related content.

Text Analytics Techniques and Use Cases

There are several techniques related to analyzing the unstructured text. Each of these techniques is used for different use case scenarios.

Sentiment analysis

Sentiment analysis is used to identify the emotions conveyed by the unstructured text. The input text includes product reviews, customer interactions, social media posts, forum discussions, or blogs. There are different types of sentiment analysis. Polarity analysis is used to identify if the text expresses positive or negative sentiment. The categorization technique is used for a more fine-grained analysis of emotions - confused, disappointed, or angry.

Use cases of sentiment analysis:

- Measure customer response to a product or a service
- Understand audience trends towards a brand
- Understand new trends in consumer space
- Prioritize customer service issues based on the severity
- Track how customer sentiment evolves over time

Topic modelling

This technique is used to find the major themes or topics in a massive volume of text or a set of documents. Topic modeling identifies the keywords used in text to identify the subject of the article.

Use cases of topic modeling:

- Large law firms use topic modeling to examine hundreds of documents during large litigations.

• Online media uses topic modeling to pick up trending topics across the web. • Researchers use topic modeling for exploratory literature review.
• Businesses can determine which of their products are successful.
• Topic modeling helps anthropologists to determine the emergent issues and trends in a society based on the content people share on the web.

Named Entity Recognition (NER)

NER is a text analytics technique used for identifying named entities like people, places, organizations, and events in unstructured text. NER extracts nouns from the text and determines the values of these nouns.

Use cases of named entity recognition:

NER is used to classify news content based on people, places, and organizations featured in them.

• Search and recommendation engines use NER for information retrieval. • For large chain companies, NER is used to sort customer service requests and assign them to a specific city, or outlet.
• Hospitals can use NER to automate the analysis of lab reports.

Term frequency – inverse document frequency

TF-IDF is used to determine how often a term appears in a large text or group of documents and therefore that term's importance to the document. This technique uses an inverse document frequency factor to filter out frequently occurring yet non-insightful words, articles, propositions, and conjunctions.

Event extraction

This is a text analytics technique that is an advancement over the named entity extraction. Event extraction recognizes events mentioned in text content, for example, mergers, acquisitions, political moves, or important meetings. Event extraction requires an advanced understanding of the semantics of text content. Advanced algorithms strive to recognize not only events but the venue, participants, date, and time wherever applicable. Event extraction is a beneficial technique that has multiple uses across fields.

Use cases of event extraction:

Link analysis: This is a technique to understand "who met whom and when" through event extraction from communication over social media. This is used by law enforcement agencies to predict possible threats to national security.

Geospatial analysis: When events are extracted along with their locations, the insights can be used to overlay them on a map. This is helpful in the geospatial analysis of the events.

Business risk monitoring: Large organizations deal with multiple partner companies and suppliers. Event extraction techniques allow businesses to monitor the web to find out if any of their partners, like suppliers or vendors, are dealing with adverse events like lawsuits or bankruptcy.

Steps Involved with Text Analytics

Text analytics is a sophisticated technique that involves several pre-steps to gather and cleanse the unstructured text. There are different ways in which text analytics can be performed. This is an example of a model workflow.

Data gathering - Text data is often scattered around the internal databases of an organization, including in customer chats, emails, product reviews, service tickets and Net Promoter Score surveys. Users also generate external data in the form of blog posts, news, reviews, social media posts and web forum discussions. While the internal data is readily available for analytics, the external data needs to be gathered.

Preparation of data - Once the unstructured text data is available, it needs to go through several preparatory steps before machine learning algorithms can analyze it. In most of the text analytics software, this step happens automatically. Text preparation includes several techniques using natural language processing as follows:

Tokenization: In this step, the text analysis algorithms break the continuous string of text data into tokens or smaller units that make up entire words or phrases. For instance, character tokens could be each individual letter in this word: F-I-S-H. Or, you can break up by subword tokens: Fish-ing. Tokens represent the basis of all natural language processing. This step also discards all the unwanted contents of the text, including white spaces.

Part-of-speech-tagging: In this step, each token in the data is assigned a grammatical category like noun, verb, adjective, and adverb.

Parsing: Parsing is the process of understanding the syntactical structure of the text. Dependency parsing and constituency parsing are two popular techniques used to derive syntactical structure.

Lemmatization and stemming: These are two processes used in data preparation to remove the suffixes and affixes associated with the tokens and retain its dictionary form or lemma.

Stopword removal: This is the phase when all the tokens that have frequent occurrence but bear no value in the text analytics. This includes words such as 'and', 'the' and 'a'. **Text analytics** - After the preparation of unstructured text data, text analytics techniques can now be performed to derive insights. There are several techniques used for text analytics. Prominent among them are text classification and text extraction.

Text classification: This technique is also known as text categorization or tagging. In this step, certain tags are assigned to the text based on its meaning. For example, while analyzing customer reviews, tags like "positive" or "negative" are assigned. Text classification often is done using rule-based systems or machine learning-based systems. In rule-based systems, humans define the association between language pattern and a tag. "Good" may indicate positive review; "bad" may idenitfy a negative review.

Machine learning systems use past examples or training data to assign tags to a new set of data. The training data and its volume are crucial, as larger sets of data helps the machine learning algorithms to give accurate tagging results. The main algorithms used in text classification are

Support Vector Machines (SVM), Naive Bayes family of algorithms (NB), and deep learning algorithms.

Text extraction: This is the process of extracting recognizable and structured information from the unstructured input text. This information includes keywords, names of people, places and events. One of the simple methods for text extraction is regular expressions. However, this is a complicated method to maintain when the complexity of input data increases. Conditional Random Fields (CRF) is a statistical method used in text extraction. CRF is a sophisticated but effective way of extracting vital information from the unstructured text.

SALES ANALYTICS:

E Commerce sales mode, sales metrics profitability metrics and support metrics.

Sales analytics is used in identifying, modeling, understanding and predicting sales trends and outcomes while aiding sales management in understanding where salespeople can improve. Specifically, sales analytic systems provide functionality that supports discovery, diagnostic and predictive exercises that enable the manipulation of parameters, measures, dimensions or figures as part of an analytic or planning exercise.

Sales analytics refers to the technology and processes used to gather sales data and gauge sales performance. Sales leaders use these metrics to set goals, improve internal processes, and forecast future sales and revenue more accurately.

The goal of sales analytics is always to simplify the information available to you. It should help you clearly understand your team's performance, sales trends, and opportunities. Generally, sales analytics is divided into four categories:

Descriptive: What happened? Descriptive analytics entails tracking historical sales data—revenue, number of users, etc.—so you can make comparisons and better understand what's currently happening.

Diagnostic: Why did it happen? Diagnostic analytics is examining and drilling down into the data to determine exactly why something occurred.

Predictive: What's going to happen? Predictive analytics is taking what you've learned about past sales and using it to gauge patterns and trends. This allows you to make educated predictions.

Prescriptive: What's the best solution or action? Prescriptive analytics involves assessing all the data and recommending the best plan of action.

Benefits of sales analytics

Sales analytics is your sales team's hidden superpower. It can enable your agents to spot key trends, dive deep, predict outcomes, and increase productivity. Accurate analysis also gives your team the ability to tailor their efforts and prioritize high-value prospects. Plus, it may even help spotlight new opportunities for your business to pursue.

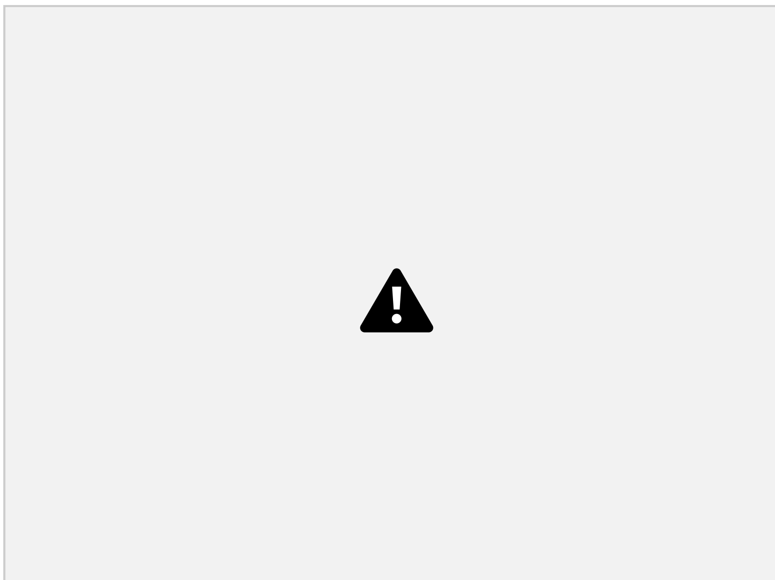
Sales analytics allows you to better gauge team performance and uncover areas for improvement, too. Understanding those strengths and weaknesses leads to better training, more attainable goals, and a cohesive team.

9 sales metrics to watch

Data is at the heart of your analytics. Before you can dive into any sales analysis, you need to understand the metrics and key performance indicators (KPIs) you're looking at and why you're measuring them. Your team can track and analyze a variety of sales metrics, including:

Sales growth

Sales growth shows how much your revenue increases (or decreases) over a specific period of time. This metric provides a bird's-eye view of sales and how your team is performing. To determine sales growth, take the sales total for the current period and subtract the sales total from the previous period. Divide that result by sales from the previous period, then multiply by 100 to get your growth percentage.



For example, say you're looking to track sales growth quarter over quarter. You had $100,000 in sales last quarter and $120,000 in sales this quarter. That means you experienced 20-percent growth in sales.

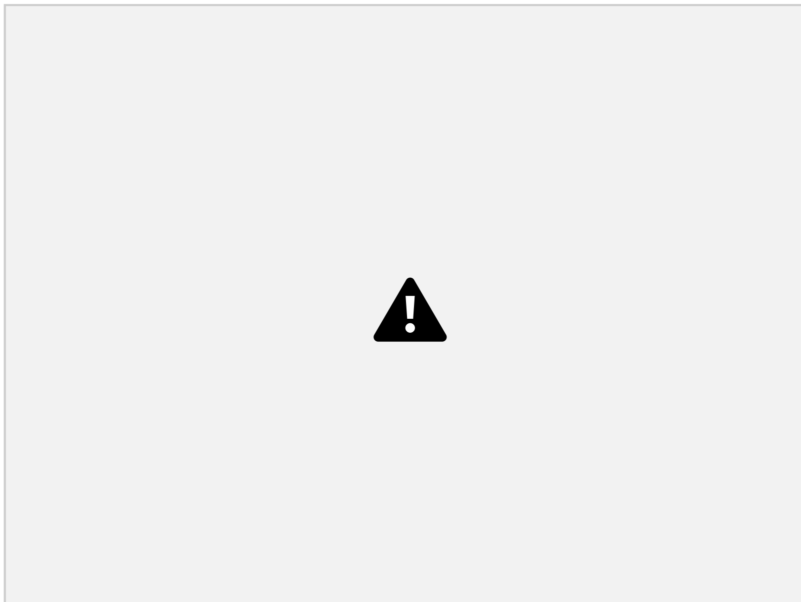
This an important metric because growth closely aligns with overall profitability. No company wants to stagnate, so it's critical to track whether or not you're experiencing healthy growth. Plus, understanding your team's performance allows for more accurate sales forecasting and more effective sales strategies.

Sales target

Sales target evaluates current sales and compares them to your bigger, long-term goals. To track this metric, you first have to determine your target. Sales targets are often based on past growth rates and revenue needed to stay in business and remain competitive. Sales targets should strike a good balance between ambitious and achievable.

Now, depending on your company and what you want to measure, your sales target can be an actual monetary value, the number of sales made, or the number of accounts opened. You can also look at various sales periods—weekly, monthly, quarterly, or yearly—to obtain the value that's most useful to you.

Once you know your sales target, it's easy to calculate the percentage. Simply divide the number of sales from the current period by the sales target, then multiply the result by 100. That number will tell you how close you are to reaching your overall target or goal.



Let's look at how this would play out. Say your company is shooting for $1 million in sales for the year. That's your sales target. Currently, you have $720,000 in sales so far for the year. That means you've reached 72 percent of your sales target, with 28 percent left to go by the end of the year.

You can also use this formula on a smaller scale. For example, say you've set a sales target for each agent to make 10 sales per month. But one agent made only six sales this month; that means they missed their monthly target by 40 percent.

It's imperative to set and monitor sales targets because they show whether the sales team is on track to meet its goals. It can also be motivating for sales agents to see how their daily efforts contribute toward larger company goals.

Sales targets can also help you determine whether the goals you've set are realistic. One sales agent missing the mark is one thing. But if most agents miss the target on an ongoing basis, then it might be time to reevaluate.

Sales per rep

Sales per rep measures the individual performance of your agents.

For this metric, all you do is track the sales each agent makes during a designated period, such as weekly, monthly, or quarterly. Again, it's good to look at both the number of deals made as well as the monetary value.

For example, maybe agent A made 10 sales and agent B made six sales. But agent A's sales totaled $100,000, while agent B's totaled $110,0000. Agent B likely sold some higher-value items or services.

It's important to look at this metric critically. Some sales representatives may have more experience or be focused on selling higher-value items, as shown in the above example. Factors like sick time, vacation, and time spent on other job duties can also influence these numbers. Sales per rep does come in handy when determining ongoing training, professional development opportunities, and rewards. Agents with lower sales numbers may benefit from mentoring or training from the sales manager, more experienced agents, or third-party sources. Sales leaders with high sales per rep may be good candidates for promotions and bonuses.

Sales by region

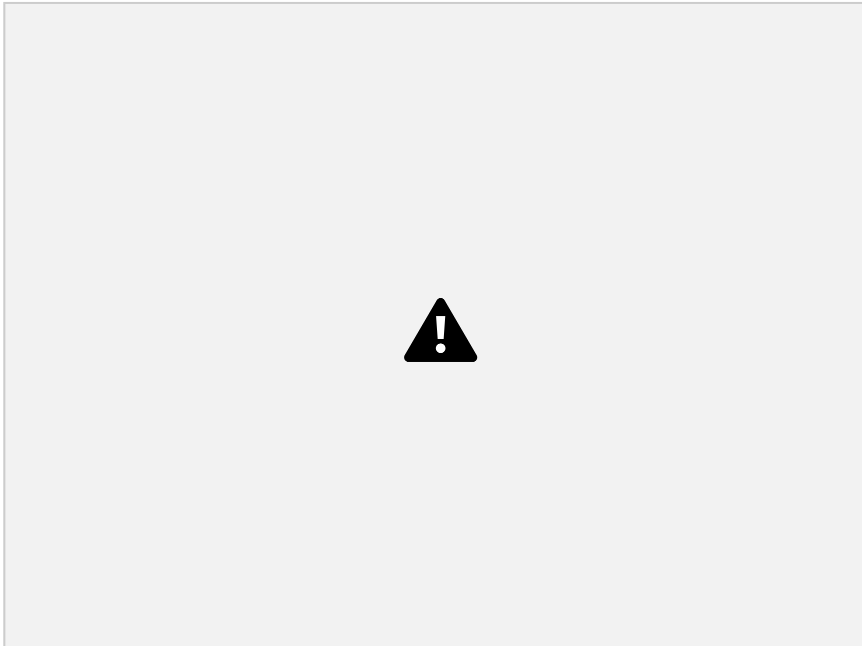
Sales by region dives into the volume of sales in key geographical areas for your business. This metric tracks sales in a specific region—territory, state, country, or continent—over a specific period of time. For example, a business that sells across the United States may want to see the annual sales generated by individual states. Maybe this year, you saw $100,000 in sales in California, $50,000 in Wisconsin, and so on.

Meanwhile, a global business may compare sales by continent. Companies that organize their sales team by territories—where rep A is in charge of sales in the northeast region, rep B focuses on the southwest region, etc.—may also want to analyze sales based on those geographic markers.

Examining sales per region can help you determine where your products or services are selling the best, enabling you to focus your sales and marketing efforts accordingly. It also reminds you to take into account certain factors (such as population density and seasonality) when setting your sales goals. For example, California is much larger and more populated than Delaware, so you'd likely aim for higher sales in California.

Sell-through rate

The sell-through rate assesses how quickly a business can sell its inventory. To calculate this metric, divide the amount of inventory you sold within a certain time period by the amount of inventory you received within that same time period. Then, multiply the result by 100 to get your sell-through rate percentage.



Say a shoe store has 100 pairs of sneakers when it receives inventory at the beginning of the month. By the end of the month, the store sells 20 pairs of sneakers. That means its sell through rate is 20 percent.

This is an important metric for retail businesses to look at when monitoring the supply chain and competitors. You want to aim for a high sell-through rate. A low sell-through rate means that your products are sitting on the shelf and not making you a profit. But running out of products isn’t good, either, so ensure you’re adjusting your inventory strategy as needed.

Sales per product

Sales per product, also called product performance, shows the profitability of each product you sell.

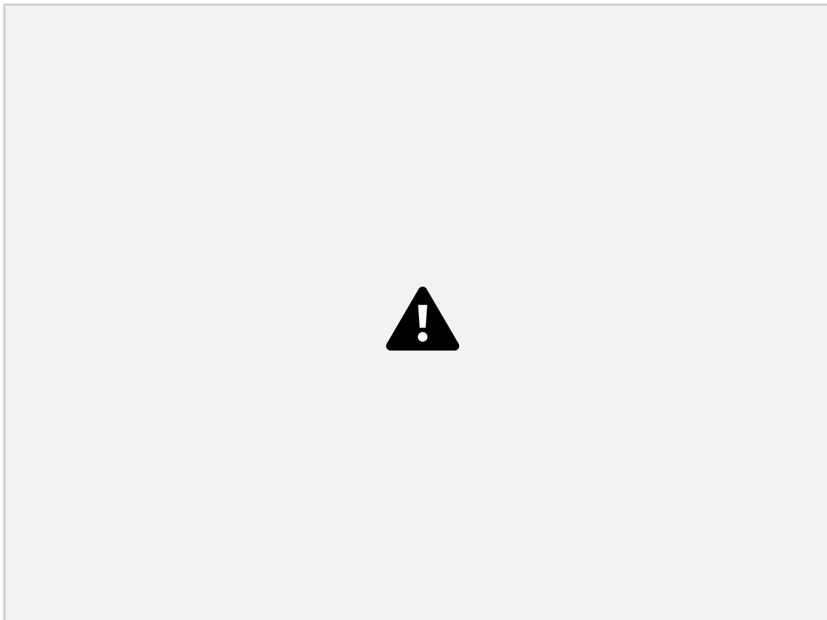
For this metric, you track total revenue for individual products over a specified period of time. Depending on your goals, you can also look at the number of units sold rather than the monetary value.

To see this play out, consider a game store that sells video games, board games, and yard games. By tracking sales per product, the store manager can see how each type of game is selling. Video games normally have a higher price point, so the store may sell fewer yet make more off each sale. Board games are generally less expensive, but the store sells a lot of them year-round, so they account for a large portion of the total sales. And lastly, yard games are a big revenue driver in the spring and summer months when the weather is nice.

Overall, this metric showcases which products are selling well and which aren’t. As depicted in the scenario above, it can help you see how high-cost, low-volume products compare to low cost, high-volume products. You can also examine how seasonality and marketing initiatives impact the sales and revenue of certain products. For instance, perhaps there’s a surge in sales on a specific product after launching a holiday promotion.

Pipeline velocity

Pipeline velocity measures how quickly leads and prospects move through your sales pipeline. To determine your pipeline velocity, look at the number of sales-qualified leads (SQLs) in your pipeline, the average deal size, and your average win rate. Multiply them together, and then divide that number by the current length of your sales cycle. This will show you how much revenue is flowing through your pipeline at any given moment.

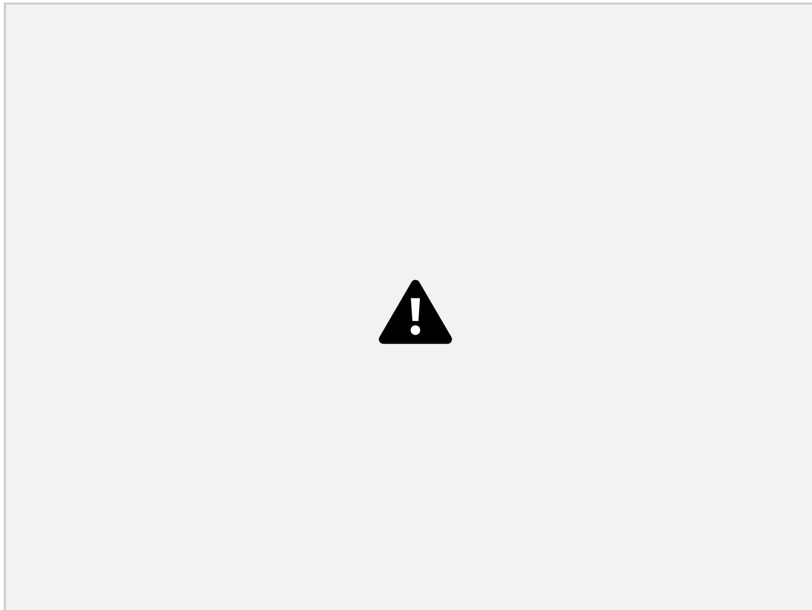


Let’s say you have 25 SQLs in your pipeline, your average deal size is about $20,000, and your average win rate is 50 percent. If your average sales cycle is about 30 days, then your current pipeline velocity is $8,333. In short, that means an estimated $8,333 is flowing through the pipeline each day.

Pipeline velocity is a key metric because you can use it for forecasting and business planning. But it’s also critical to work through what might occur if just one of those factors changed: What would happen if your SQLs were suddenly cut in half for the month? Or your sales cycle doubled? Or you started winning 75 percent of your deals? Any change can have a dramatic effect on your pipeline velocity, which can impact goals and initiatives for your team.

Quote to close

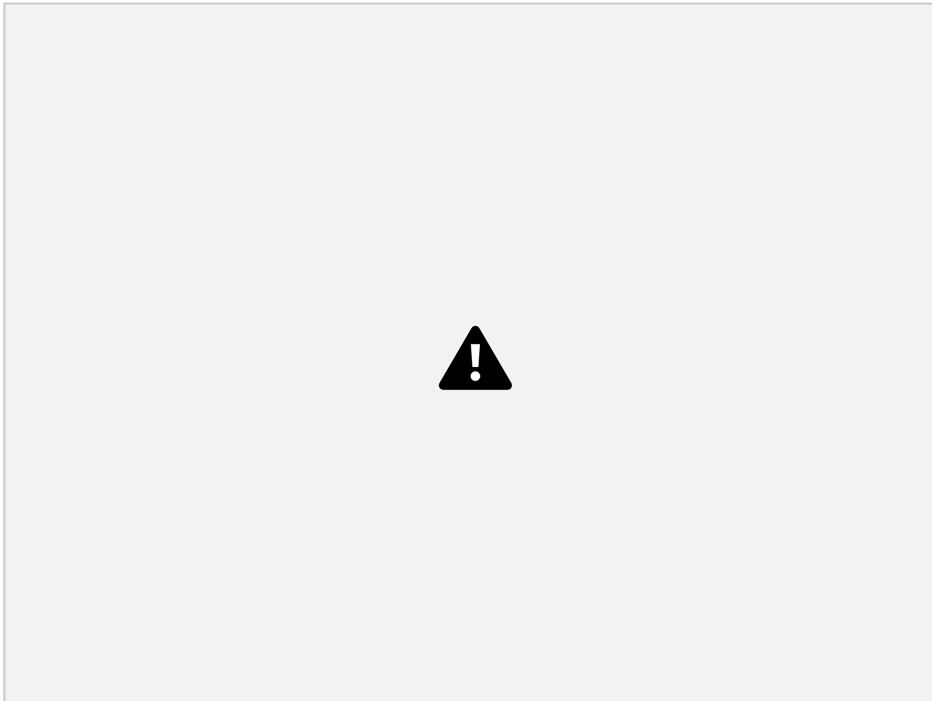
Quote to close determines the percentage of prospects who turn into paying customers. To determine your quote-to-close rate, take the number of actual deals closed during a specific time period and divide it by the number of quotes sent during that same time period. Then, multiply the result by 100 to get your quote-to-close percentage.



Imagine you sent out 100 quotes to potential customers over the past year. Of those, you closed 60 of them. That means you have a 60 percent quote-to-close rate. This metric matters because it illustrates how well your team can move prospects through the sales funnel and turn them into customers. If the number is steadily decreasing or relatively low, then you might need to make adjustments to your pricing or your ideal customer profile (ICP). It may also signal more training opportunities for your team.

Average purchase value

Average purchase value, or average sale value, examines the average value of each transaction. To calculate this metric, take your total sales revenue and divide it by the number of sales you made.



Say a jewelry store made $2 million in sales last year. There were 1,000 purchases. That means the average purchase value was around $2,000.

Average purchase value is helpful when making sales forecasts and projections, as you can quickly estimate how many customers and/or purchases you'd need to reach a lofty revenue goal. You can then use it to plan upsell or marketing strategies.

If you were looking to boost the average purchase value, you could prioritize marketing, selling higher-ticket items, or pushing upsell opportunities. In the case of a jewelry store, that might mean marketing your larger stones or more expensive settings and upselling an extended warranty option.

What to look for in sales analytics tools

There are various sales analytics tools available to help you track these metrics. But not all tools are created equal. There are a few specific features you should look for when choosing a sales analytics solution.

Interactive visualization

Your sales analytics tool should turn the data into an easy-to-digest format. As they say, a picture is worth a thousand words, so the more visual you can make the data, the better. Look for visual dashboards that allow your sales teams and company leaders to view sales activities at a glance. You should also find a solution that's interactive and allows teams to drag, drop, and adjust the data as needed.

A visual and interactive analytics tool helps your sales team tell a story with the numbers and make quick, data-driven decisions. Plus, it keeps the focus on high-value leads and actionable insights without getting lost in the numbers.

Ease of use

You want the tool to have features that reduce friction and make it simple for sales teams to focus on selling rather than create more work. Features like automatic syncing and real-time updates help save time and keep information current without requiring additional steps for your team. For example, if a customer support agent documents a ticket, it should automatically sync with the customer's account so the sales team is up-to-speed on any issues. Similarly, sales dashboards should update automatically when new sales are made or goals are met. Mobile capabilities are handy, too, because sales teams are often remote or out on the road meeting with customers. That way, they can reference account information and make updates to customer data without having to wait until they're back in the office.

Integration capabilities

Data comes from a variety of places, so your sales analytics solution should work seamlessly alongside the other tools your team regularly uses.

Look for tools that integrate with existing apps—such as your email marketing platform, calendar, and task management software. This will simplify workflows and save your sales team valuable time.

Of course, every company is different, so seek customization opportunities to meet your specific needs. That may mean pulling specific metrics or building your own application programming interface.